Project Assignment 2: Use Case Implementation with Design Pattern Integration

Objective:

For this assignment, each group should select one use case from their project and implement it. The selected use case must demonstrate the design patterns previously mapped to it. Submit your implementation to GitHub and include a task matrix (TASK_MATRIX.md) in your repository that outlines the workload distribution—indicating which team member contributed to which parts of the assignment. Prepare a software design document describing the implemented software's design.

Deadline: 1 July 2025, 23:59

Submission Instructions:

- Each team is required to submit their work on GitHub, ensuring clear commit messages and contribution tracking for each member.
- Maintain the traceability via numbering and naming conventions throughout the design, implementation and test for the selected use case.

Deliverables:

- Software Design Document
- Source code of the selected use case

GitHub Submission: You will submit your work for each assignment through GitHub.

README: Include a Readme.md file in the project to guide the reviewer about the project.

Recommended Folder Structure:

/project-root

/src

/docs

SoftwareDesignDocument.pdf

TASK_MATRIX.md

README.md

Individual GitHub Accounts: Each team member must push their contributions using their own GitHub account within the shared team repository.

Clarity of Contribution: On GitHub, it should be evident which team member is working on which aspects of the project. Use clear commit messages and file structures to show individual contributions.

Document Authorship: On the first page of each document, please list the names of the team members who contributed to the creation of that specific document

Comments: In your code, add meaningful comments where each design pattern is applied. Each comment should include the name of the design pattern, the intent behind its use, the motivation for selecting that pattern, and how it improves the implementation in this context. Where possible, provide measurable metrics that reflect the improvement enabled by the pattern's application.

Document-Specific Task Matrix: Within each document you submit, you are required to include a document-specific task matrix. This matrix should detail the tasks involved in creating that specific document and clearly show which team member was responsible for each task within the document's creation process. This helps illustrate the division of work within your team for each deliverable

Software Design Document:

Discuss and explain the following subjects in this document for the selected use case. Provide UML diagrams when necessary.

- System Overview
- System Context
- Key Features and Functionality
- Assumptions and Dependencies
- Architectural Design
    - System Architecture Diagram (High-Level)
    - Architectural Patterns and Styles (e.g., Microservices, MVC, Layered)
    - Rationale for Architectural Decisions
- Component Design
    - Subsystems and Modules
    - Responsibilities of Each Component
    - Interfaces Between Components
    - Component Diagrams
- Data Design
    - Data Model / ER Diagram
    - Data Storage (Database or File Structure)
    - Data Flow Diagrams
    - Data Validation Rules
- Design Patterns
    - Applied Design Patterns
    - Context and Justification
- Implementation Notes
- User Interface Design (if applicable)
    - UI Mockups or Wireframes
- External Interfaces

    - APIs
    - Third-party Systems
- Performance Considerations
    - Performance Requirements
    - Scalability & Optimization Strategies
- Error Handling and Logging
    - Exception Management
    - Logging Mechanisms
- Design for Testability
- Deployment and Installation Design
    - Environment Configuration
    - Packaging and Dependencies
- Change Log
- Future Work / Open Issues